

Subscription & Billing Management Platform

MySQL DDL Script — All Tables with Column Reference

Version 1.0 | March 2026 | MySQL 8.0+

Overview

This document contains the complete MySQL DDL script for the Subscription & Billing Management Platform. All 17 tables are organised by functional domain. For each table, a column reference table is provided followed by the exact CREATE TABLE statement ready to execute against a MySQL 8.0+ database.

Tables must be created in the order they appear in this document. Foreign key constraints reference tables defined earlier in the script, so creation order is significant.

Creation Order Summary

Execute tables in this order to satisfy all foreign key dependencies.

#	Table	Dependencies
1	user	No dependencies
2	customer	No dependencies
3	audit_log	Depends on: user
4	plan	No dependencies
5	plan_feature	Depends on: plan
6	discount	No dependencies
7	subscription	Depends on: customer, plan
8	subscription_addon	Depends on: subscription
9	subscription_discount	Depends on: subscription, discount
10	tax_rule	No dependencies
11	billing_job	No dependencies
12	invoice	Depends on: subscription, billing_job
13	invoice_line_item	Depends on: invoice
14	invoice_tax	Depends on: invoice, tax_rule
15	payment_method	Depends on: customer
16	payment	Depends on: invoice, payment_method
17	dunning_event	Depends on: invoice

Domain 1 — Customer & Identity Management

Tables that manage external customers, internal staff access, and the immutable audit trail. These are the foundation of RBAC and compliance.

user

Internal staff accounts. Drives role-based access control (RBAC) across all platform operations.

Column reference

Column	Type	Notes / Constraints
id	CHAR(36)	Primary key — UUID generated by MySQL
name	VARCHAR(150)	Staff member full name — NOT NULL
email	VARCHAR(255)	Login email — UNIQUE constraint
password_hash	VARCHAR(255)	Bcrypt / Argon2 hash — never plain text
role	ENUM	admin finance support readonly
is_active	BOOLEAN	Soft disable without deletion — default TRUE
last_login	DATETIME	NULL until first login
created_at	DATETIME	Auto-set on INSERT via DEFAULT CURRENT_TIMESTAMP

CREATE TABLE statement

```
CREATE TABLE user (  
  id          CHAR(36)      NOT NULL DEFAULT (UUID()),  
  name        VARCHAR(150) NOT NULL,  
  email       VARCHAR(255) NOT NULL,  
  password_hash VARCHAR(255) NOT NULL,  
  role        ENUM('admin','finance','support','readonly')  
              NOT NULL DEFAULT 'readonly',  
  is_active   BOOLEAN      NOT NULL DEFAULT TRUE,  
  last_login  DATETIME      NULL,  
  created_at  DATETIME      NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  CONSTRAINT pk_user      PRIMARY KEY (id),  
  CONSTRAINT uq_user_email UNIQUE (email)  
);
```

customer

Central registry of all external customers. Every subscription, invoice, and payment method traces back here.

Column reference

Column	Type	Notes / Constraints
id	CHAR(36)	Primary key — UUID
name	VARCHAR(150)	Customer or contact name
email	VARCHAR(255)	Billing email — UNIQUE constraint
phone	VARCHAR(30)	Optional contact number
company	VARCHAR(150)	Business name — nullable
billing_address	TEXT	Full address printed on invoices
tax_id	VARCHAR(50)	VAT / GST / PAN number — nullable
status	ENUM	active suspended churned trial
created_at	DATETIME	Account creation timestamp
updated_at	DATETIME	Auto-updated on every row change

CREATE TABLE statement

```
CREATE TABLE customer (
  id          CHAR(36)      NOT NULL DEFAULT (UUID()),
  name        VARCHAR(150) NOT NULL,
  email       VARCHAR(255) NOT NULL,
  phone       VARCHAR(30)   NULL,
  company     VARCHAR(150) NULL,
  billing_address TEXT      NULL,
  tax_id      VARCHAR(50)   NULL,
  status      ENUM('active','suspended','churned','trial')
              NOT NULL DEFAULT 'active',
  created_at  DATETIME      NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at  DATETIME      NOT NULL DEFAULT CURRENT_TIMESTAMP
              ON UPDATE CURRENT_TIMESTAMP,
  CONSTRAINT pk_customer PRIMARY KEY (id),
  CONSTRAINT uq_customer_email UNIQUE (email)
);
```

audit_log

Immutable record of every sensitive action performed by internal users. Powers compliance reporting and dispute resolution.

Column reference

Column	Type	Notes / Constraints
id	CHAR(36)	Primary key — UUID
user_id	CHAR(36) FK	References user — RESTRICT on delete
entity_type	VARCHAR(50)	E.g. INVOICE, PAYMENT, SUBSCRIPTION
entity_id	CHAR(36)	ID of the affected record

Column	Type	Notes / Constraints
action	ENUM	CREATE UPDATE DELETE BILLING_RUN
old_values	JSON	Pre-change snapshot — nullable
new_values	JSON	Post-change snapshot — nullable
ip_address	VARCHAR(45)	IPv4 or IPv6 — nullable
created_at	DATETIME	Immutable timestamp of the event

CREATE TABLE statement

```
CREATE TABLE audit_log (
  id          CHAR(36)      NOT NULL DEFAULT (UUID()),
  user_id     CHAR(36)      NOT NULL,
  entity_type VARCHAR(50)   NOT NULL,
  entity_id   CHAR(36)      NOT NULL,
  action      ENUM('CREATE','UPDATE','DELETE','BILLING_RUN') NOT NULL,
  old_values  JSON          NULL,
  new_values  JSON          NULL,
  ip_address  VARCHAR(45)   NULL,
  created_at  DATETIME      NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT pk_audit_log PRIMARY KEY (id),
  CONSTRAINT fk_audit_user FOREIGN KEY (user_id)
    REFERENCES user(id) ON DELETE RESTRICT ON UPDATE CASCADE,
  INDEX idx_audit_entity (entity_type, entity_id),
  INDEX idx_audit_user   (user_id),
  INDEX idx_audit_created (created_at)
);
```

Domain 2 — Plan & Subscription Management

Tables that define what is sold (plans, features, discounts) and track the lifecycle of each customer's subscription.

plan

Defines all available subscription tiers including pricing, billing cadence, and trial configuration.

Column reference

Column	Type	Notes / Constraints
id	CHAR(36)	Primary key — UUID
name	VARCHAR(100)	Display name e.g. Pro, Enterprise — UNIQUE
description	TEXT	Marketing description — nullable
base_price	DECIMAL(12,2)	Price per billing cycle — CHECK >= 0
billing_cycle	ENUM	monthly quarterly annual usage
plan_type	ENUM	flat per_seat usage_based tiered
trial_days	INT	Free trial days — CHECK >= 0, default 0
is_active	BOOLEAN	FALSE hides plan from new signups
created_at	DATETIME	Plan creation timestamp

CREATE TABLE statement

```
CREATE TABLE plan (  
  id          CHAR(36)          NOT NULL DEFAULT (UUID()),  
  name        VARCHAR(100)      NOT NULL,  
  description  TEXT              NULL,  
  base_price   DECIMAL(12, 2)    NOT NULL DEFAULT 0.00,  
  billing_cycle ENUM('monthly','quarterly','annual','usage')  
              NOT NULL DEFAULT 'monthly',  
  plan_type    ENUM('flat','per_seat','usage_based','tiered')  
              NOT NULL DEFAULT 'flat',  
  trial_days   INT              NOT NULL DEFAULT 0,  
  is_active    BOOLEAN          NOT NULL DEFAULT TRUE,  
  created_at   DATETIME         NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  CONSTRAINT pk_plan PRIMARY KEY (id),  
  CONSTRAINT uq_plan_name UNIQUE (name),  
  CONSTRAINT chk_plan_price CHECK (base_price >= 0),  
  CONSTRAINT chk_plan_trial CHECK (trial_days >= 0)  
);
```

plan_feature

Feature entitlements tied to each plan. Used for feature-gating and pricing page comparison tables.

Column reference

Column	Type	Notes / Constraints
id	CHAR(36)	Primary key — UUID
plan_id	CHAR(36) FK	References plan — CASCADE on delete
feature_name	VARCHAR(100)	E.g. api_calls, storage_gb, seats
feature_value	VARCHAR(100)	Limit or entitlement value as string
unit	VARCHAR(50)	Unit of measure e.g. calls, GB — nullable

CREATE TABLE statement

```
CREATE TABLE plan_feature (  
  id          CHAR(36)      NOT NULL DEFAULT (UUID()),  
  plan_id     CHAR(36)      NOT NULL,  
  feature_name VARCHAR(100) NOT NULL,  
  feature_value VARCHAR(100) NOT NULL,  
  unit        VARCHAR(50)   NULL,  
  CONSTRAINT pk_plan_feature PRIMARY KEY (id),  
  CONSTRAINT fk_plan_feature_plan FOREIGN KEY (plan_id)  
    REFERENCES plan(id) ON DELETE CASCADE ON UPDATE CASCADE,  
  INDEX idx_plan_feature_plan (plan_id)  
);
```

discount

Master coupon registry. Defines discount rules — flat or percentage — with time limits and usage caps.

Column reference

Column	Type	Notes / Constraints
id	CHAR(36)	Primary key — UUID
code	VARCHAR(50)	Human-readable coupon code — UNIQUE
discount_type	ENUM	flat percentage
value	DECIMAL(10,2)	Amount or percentage — CHECK > 0
valid_from	DATE	Activation date
valid_until	DATE	Expiry date — nullable means no expiry
max_uses	INT	Redemption cap — NULL means unlimited
used_count	INT	Current redemption count — default 0
is_active	BOOLEAN	Soft disable flag — default TRUE

CREATE TABLE statement

```
CREATE TABLE discount (
  id          CHAR(36)          NOT NULL DEFAULT (UUID()),
  code        VARCHAR(50)       NOT NULL,
  discount_type ENUM('flat','percentage') NOT NULL DEFAULT 'percentage',
  value       DECIMAL(10, 2)    NOT NULL,
  valid_from  DATE              NOT NULL,
  valid_until DATE              NULL,
  max_uses    INT               NULL,
  used_count  INT               NOT NULL DEFAULT 0,
  is_active   BOOLEAN           NOT NULL DEFAULT TRUE,
  CONSTRAINT pk_discount PRIMARY KEY (id),
  CONSTRAINT uq_discount_code UNIQUE (code),
  CONSTRAINT chk_discount_value CHECK (value > 0),
  CONSTRAINT chk_discount_dates CHECK
    (valid_until IS NULL OR valid_until >= valid_from),
  INDEX idx_discount_code (code),
  INDEX idx_discount_active (is_active)
);
```

subscription

Lifecycle record for each customer-plan relationship. The central hub queried by the billing engine on every run.

Column reference

Column	Type	Notes / Constraints
id	CHAR(36)	Primary key — UUID
customer_id	CHAR(36) FK	References customer — RESTRICT on delete
plan_id	CHAR(36) FK	References plan — RESTRICT on delete
status	ENUM	active trial paused cancelled past_due
start_date	DATE	Activation date
end_date	DATE	Cancellation or expiry date — nullable
trial_end_date	DATE	Computed from plan.trial_days — nullable
next_billing_date	DATE	Primary field used by billing job daily query
current_mrr	DECIMAL(12,2)	Monthly recurring revenue — CHECK >= 0
quantity	INT	Seats or units — CHECK > 0, default 1
created_at	DATETIME	Record creation timestamp
updated_at	DATETIME	Auto-updated on status change

CREATE TABLE statement

```
CREATE TABLE subscription (
  id          CHAR(36)          NOT NULL DEFAULT (UUID()),
  customer_id CHAR(36)          NOT NULL,
  plan_id     CHAR(36)          NOT NULL,
  status      ENUM('active','trial','paused','cancelled','past_due')
```

```

start_date          DATE          NOT NULL DEFAULT 'trial',
end_date            DATE          NOT NULL,
trial_end_date      DATE          NULL,
next_billing_date   DATE          NOT NULL,
current_mrr         DECIMAL(12, 2) NOT NULL DEFAULT 0.00,
quantity            INT           NOT NULL DEFAULT 1,
created_at          DATETIME      NOT NULL DEFAULT CURRENT_TIMESTAMP,
updated_at          DATETIME      NOT NULL DEFAULT CURRENT_TIMESTAMP
                                ON UPDATE CURRENT_TIMESTAMP,
CONSTRAINT pk_subscription PRIMARY KEY (id),
CONSTRAINT fk_subscription_customer FOREIGN KEY (customer_id)
    REFERENCES customer(id) ON DELETE RESTRICT ON UPDATE CASCADE,
CONSTRAINT fk_subscription_plan FOREIGN KEY (plan_id)
    REFERENCES plan(id) ON DELETE RESTRICT ON UPDATE CASCADE,
CONSTRAINT chk_subscription_quantity CHECK (quantity > 0),
CONSTRAINT chk_subscription_mrr CHECK (current_mrr >= 0),
INDEX idx_subscription_customer (customer_id),
INDEX idx_subscription_plan (plan_id),
INDEX idx_subscription_status (status),
INDEX idx_subscription_next_billing (next_billing_date)
);

```

subscription_addon

Optional add-on charges purchased on top of a base plan (e.g. extra storage, priority support).

Column reference

Column	Type	Notes / Constraints
id	CHAR(36)	Primary key — UUID
subscription_id	CHAR(36) FK	References subscription — CASCADE on delete
addon_name	VARCHAR(100)	Name of the add-on service
price	DECIMAL(12,2)	Price per billing cycle — CHECK >= 0
quantity	INT	Number of add-on units — CHECK > 0

CREATE TABLE statement

```

CREATE TABLE subscription_addon (
  id          CHAR(36)          NOT NULL DEFAULT (UUID()),
  subscription_id CHAR(36)      NOT NULL,
  addon_name   VARCHAR(100)     NOT NULL,
  price        DECIMAL(12, 2)   NOT NULL DEFAULT 0.00,
  quantity     INT              NOT NULL DEFAULT 1,
  CONSTRAINT pk_subscription_addon PRIMARY KEY (id),
  CONSTRAINT fk_addon_subscription FOREIGN KEY (subscription_id)
    REFERENCES subscription(id) ON DELETE CASCADE ON UPDATE CASCADE,
  CONSTRAINT chk_addon_price CHECK (price >= 0),
  CONSTRAINT chk_addon_quantity CHECK (quantity > 0),
  INDEX idx_addon_subscription (subscription_id)
);

```


subscription_discount

Junction table linking discounts to subscriptions. Tracks when a discount was applied and when it expires.

Column reference

Column	Type	Notes / Constraints
id	CHAR(36)	Primary key — UUID
subscription_id	CHAR(36) FK	References subscription — CASCADE on delete
discount_id	CHAR(36) FK	References discount — RESTRICT on delete
applied_at	DATE	Date the discount was first applied
expires_at	DATE	Date the discount stops applying — nullable

CREATE TABLE statement

```
CREATE TABLE subscription_discount (  
  id CHAR(36) NOT NULL DEFAULT (UUID()),  
  subscription_id CHAR(36) NOT NULL,  
  discount_id CHAR(36) NOT NULL,  
  applied_at DATE NOT NULL DEFAULT (CURRENT_DATE),  
  expires_at DATE NULL,  
  CONSTRAINT pk_subscription_discount PRIMARY KEY (id),  
  CONSTRAINT fk_sub_discount_subscription FOREIGN KEY (subscription_id)  
    REFERENCES subscription(id) ON DELETE CASCADE ON UPDATE CASCADE,  
  CONSTRAINT fk_sub_discount_discount FOREIGN KEY (discount_id)  
    REFERENCES discount(id) ON DELETE RESTRICT ON UPDATE CASCADE,  
  CONSTRAINT uq_sub_discount UNIQUE (subscription_id, discount_id),  
  INDEX idx_sub_discount_sub (subscription_id),  
  INDEX idx_sub_discount_disc (discount_id)  
);
```

Domain 3 — Billing Engine

Tables that power the automated billing pipeline — tax configuration, daily job execution, invoices, and itemised line items.

tax_rule

Region-based tax configuration. The billing engine matches a customer's billing address region to apply the correct rate on invoices.

Column reference

Column	Type	Notes / Constraints
id	CHAR(36)	Primary key — UUID
name	VARCHAR(100)	Rule name e.g. Karnataka GST 18%
region	VARCHAR(20)	Country or state code e.g. IN-KA, US-CA
rate	DECIMAL(6,4)	E.g. 0.1800 for 18% — CHECK 0 to 1
tax_type	ENUM	GST VAT sales_tax service_tax
is_active	BOOLEAN	FALSE preserves old rule for audit history

CREATE TABLE statement

```
CREATE TABLE tax_rule (  
  id          CHAR(36)      NOT NULL DEFAULT (UUID()),  
  name        VARCHAR(100)  NOT NULL,  
  region      VARCHAR(20)   NOT NULL,  
  rate        DECIMAL(6, 4) NOT NULL,  
  tax_type    ENUM('GST','VAT','sales_tax','service_tax') NOT NULL,  
  is_active   BOOLEAN       NOT NULL DEFAULT TRUE,  
  CONSTRAINT pk_tax_rule PRIMARY KEY (id),  
  CONSTRAINT chk_tax_rate CHECK (rate >= 0 AND rate <= 1),  
  INDEX idx_tax_rule_region (region),  
  INDEX idx_tax_rule_active (is_active)  
);
```

billing_job

Represents a scheduled automated billing run. Tracks execution status, volume processed, and any errors encountered.

Column reference

Column	Type	Notes / Constraints
id	CHAR(36)	Primary key — UUID

Column	Type	Notes / Constraints
scheduled_date	DATE	Date the job was due to run — UNIQUE
status	ENUM	pending running completed failed
subscriptions_processed	INT	Count of subscriptions evaluated
invoices_generated	INT	Count of invoices created in this run
total_amount	DECIMAL(15,2)	Sum of all invoices in this run
started_at	DATETIME	Actual execution start — nullable
completed_at	DATETIME	Execution end — nullable
error_log	TEXT	Failure details — nullable

CREATE TABLE statement

```
CREATE TABLE billing_job (
  id CHAR(36) NOT NULL DEFAULT (UUID()),
  scheduled_date DATE NOT NULL,
  status ENUM('pending','running','completed','failed') NOT NULL DEFAULT 'pending',
  subscriptions_processed INT NOT NULL DEFAULT 0,
  invoices_generated INT NOT NULL DEFAULT 0,
  total_amount DECIMAL(15, 2) NOT NULL DEFAULT 0.00,
  started_at DATETIME NULL,
  completed_at DATETIME NULL,
  error_log TEXT NULL,
  CONSTRAINT pk_billing_job PRIMARY KEY (id),
  CONSTRAINT uq_billing_job_date UNIQUE (scheduled_date),
  INDEX idx_billing_job_status (status),
  INDEX idx_billing_job_date (scheduled_date)
);
```

invoice

Authoritative financial document generated per billing cycle per subscription. Drives payment collection and dunning workflows.

Column reference

Column	Type	Notes / Constraints
id	CHAR(36)	Primary key — UUID
subscription_id	CHAR(36) FK	References subscription — RESTRICT on delete
billing_job_id	CHAR(36) FK	References billing_job — SET NULL on delete
invoice_number	VARCHAR(50)	Human-readable reference e.g. INV-2024-00123 — UNIQUE
issue_date	DATE	Date invoice was generated

Column	Type	Notes / Constraints
due_date	DATE	Payment deadline — CHECK >= issue_date
subtotal	DECIMAL(12,2)	Sum before discounts and taxes
discount_amount	DECIMAL(12,2)	Total discount applied
tax_amount	DECIMAL(12,2)	Total tax applied
total_amount	DECIMAL(12,2)	Final payable amount
status	ENUM	draft sent paid overdue void
notes	TEXT	Additional billing notes — nullable
created_at	DATETIME	Invoice creation timestamp

CREATE TABLE statement

```
CREATE TABLE invoice (
  id CHAR(36) NOT NULL DEFAULT (UUID()),
  subscription_id CHAR(36) NOT NULL,
  billing_job_id CHAR(36) NULL,
  invoice_number VARCHAR(50) NOT NULL,
  issue_date DATE NOT NULL,
  due_date DATE NOT NULL,
  subtotal DECIMAL(12, 2) NOT NULL DEFAULT 0.00,
  discount_amount DECIMAL(12, 2) NOT NULL DEFAULT 0.00,
  tax_amount DECIMAL(12, 2) NOT NULL DEFAULT 0.00,
  total_amount DECIMAL(12, 2) NOT NULL DEFAULT 0.00,
  status ENUM('draft','sent','paid','overdue','void')
    NOT NULL DEFAULT 'draft',
  notes TEXT NULL,
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT pk_invoice PRIMARY KEY (id),
  CONSTRAINT uq_invoice_number UNIQUE (invoice_number),
  CONSTRAINT fk_invoice_sub FOREIGN KEY (subscription_id)
    REFERENCES subscription(id) ON DELETE RESTRICT ON UPDATE CASCADE,
  CONSTRAINT fk_invoice_billing FOREIGN KEY (billing_job_id)
    REFERENCES billing_job(id) ON DELETE SET NULL ON UPDATE CASCADE,
  CONSTRAINT chk_invoice_amounts CHECK (
    subtotal >= 0 AND discount_amount >= 0
    AND tax_amount >= 0 AND total_amount >= 0),
  CONSTRAINT chk_invoice_dates CHECK (due_date >= issue_date),
  INDEX idx_invoice_subscription (subscription_id),
  INDEX idx_invoice_billing_job (billing_job_id),
  INDEX idx_invoice_status (status),
  INDEX idx_invoice_due_date (due_date)
);
```

invoice_line_item

Itemised breakdown within an invoice. Each plan charge, add-on, or adjustment is a separate line.

Column reference

Column	Type	Notes / Constraints
id	CHAR(36)	Primary key — UUID
invoice_id	CHAR(36) FK	References invoice — CASCADE on delete
description	VARCHAR(255)	Charge label e.g. Pro Plan — October 2024
quantity	INT	Units billed — CHECK > 0
unit_price	DECIMAL(12,2)	Price per unit — CHECK >= 0
discount_amount	DECIMAL(12,2)	Discount on this line — CHECK >= 0
line_total	DECIMAL(12,2)	qty * unit_price - discount — CHECK >= 0

CREATE TABLE statement

```
CREATE TABLE invoice_line_item (
  id          CHAR(36)          NOT NULL DEFAULT (UUID()),
  invoice_id  CHAR(36)          NOT NULL,
  description VARCHAR(255)      NOT NULL,
  quantity    INT               NOT NULL DEFAULT 1,
  unit_price  DECIMAL(12, 2)    NOT NULL DEFAULT 0.00,
  discount_amount DECIMAL(12, 2) NOT NULL DEFAULT 0.00,
  line_total  DECIMAL(12, 2)    NOT NULL DEFAULT 0.00,
  CONSTRAINT pk_invoice_line_item PRIMARY KEY (id),
  CONSTRAINT fk_line_item_invoice FOREIGN KEY (invoice_id)
    REFERENCES invoice(id) ON DELETE CASCADE ON UPDATE CASCADE,
  CONSTRAINT chk_line_item_quantity CHECK (quantity > 0),
  CONSTRAINT chk_line_item_amounts CHECK (
    unit_price >= 0 AND discount_amount >= 0 AND line_total >= 0),
  INDEX idx_line_item_invoice (invoice_id)
);
```

invoice_tax

Junction table resolving the many-to-many between tax_rule and invoice. Snapshots the rate at billing time to preserve historical accuracy.

Column reference

Column	Type	Notes / Constraints
id	CHAR(36)	Primary key — UUID
invoice_id	CHAR(36) FK	References invoice — CASCADE on delete
tax_rule_id	CHAR(36) FK	References tax_rule — RESTRICT on delete
rate	DECIMAL(6,4)	Rate snapshot at invoice time
tax_amount	DECIMAL(12,2)	Calculated tax for this rule on this invoice

CREATE TABLE statement

```
CREATE TABLE invoice_tax (  
  id          CHAR(36)          NOT NULL DEFAULT (UUID()),  
  invoice_id  CHAR(36)          NOT NULL,  
  tax_rule_id CHAR(36)          NOT NULL,  
  rate        DECIMAL(6, 4)     NOT NULL,  
  tax_amount  DECIMAL(12, 2)    NOT NULL,  
  CONSTRAINT pk_invoice_tax      PRIMARY KEY (id),  
  CONSTRAINT fk_invoice_tax_invoice FOREIGN KEY (invoice_id)  
    REFERENCES invoice(id) ON DELETE CASCADE,  
  CONSTRAINT fk_invoice_tax_rule  FOREIGN KEY (tax_rule_id)  
    REFERENCES tax_rule(id) ON DELETE RESTRICT,  
  CONSTRAINT uq_invoice_tax UNIQUE (invoice_id, tax_rule_id),  
  INDEX idx_invoice_tax_invoice (invoice_id)  
);
```

Domain 4 — Payments & Collections

Tables that handle tokenised payment instruments, transaction records, and the automated dunning (retry and reminder) workflow for overdue invoices.

payment_method

Stores tokenised payment instruments per customer. Never holds raw card numbers — only the gateway vault token and last four digits for display.

Column reference

Column	Type	Notes / Constraints
id	CHAR(36)	Primary key — UUID
customer_id	CHAR(36) FK	References customer — RESTRICT on delete
method_type	ENUM	card bank_transfer upi wallet
provider_token	VARCHAR(255)	Payment gateway vault reference
last_four	CHAR(4)	Last 4 digits of card — display only, nullable
expiry_date	DATE	Card expiry — nullable for non-card methods
is_default	BOOLEAN	TRUE = charged during automated billing runs
is_active	BOOLEAN	FALSE = card expired or removed by customer
created_at	DATETIME	Record creation timestamp

CREATE TABLE statement

```
CREATE TABLE payment_method (  
  id CHAR(36) NOT NULL DEFAULT (UUID()),  
  customer_id CHAR(36) NOT NULL,  
  method_type ENUM('card','bank_transfer','upi','wallet')  
    NOT NULL DEFAULT 'card',  
  provider_token VARCHAR(255) NOT NULL,  
  last_four CHAR(4) NULL,  
  expiry_date DATE NULL,  
  is_default BOOLEAN NOT NULL DEFAULT FALSE,  
  is_active BOOLEAN NOT NULL DEFAULT TRUE,  
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  CONSTRAINT pk_payment_method PRIMARY KEY (id),  
  CONSTRAINT fk_payment_method_customer FOREIGN KEY (customer_id)  
    REFERENCES customer(id) ON DELETE RESTRICT ON UPDATE CASCADE,  
  INDEX idx_payment_method_customer (customer_id),  
  INDEX idx_payment_method_default (customer_id, is_default)  
);
```

payment

Records every payment transaction against an invoice, including gateway response for bank reconciliation.

Column reference

Column	Type	Notes / Constraints
id	CHAR(36)	Primary key — UUID
invoice_id	CHAR(36) FK	References invoice — RESTRICT on delete
payment_method_id	CHAR(36) FK	References payment_method — SET NULL on delete
amount	DECIMAL(12,2)	Amount paid — CHECK > 0
status	ENUM	pending success failed refunded
transaction_id	VARCHAR(255)	Gateway transaction reference — nullable
gateway_response	TEXT	Raw gateway payload for reconciliation
paid_at	DATETIME	Exact payment confirmation time — nullable
created_at	DATETIME	Record creation timestamp

CREATE TABLE statement

```
CREATE TABLE payment (
  id CHAR(36) NOT NULL DEFAULT (UUID()),
  invoice_id CHAR(36) NOT NULL,
  payment_method_id CHAR(36) NULL,
  amount DECIMAL(12, 2) NOT NULL,
  status ENUM('pending','success','failed','refunded')
  NOT NULL DEFAULT 'pending',
  transaction_id VARCHAR(255) NULL,
  gateway_response TEXT NULL,
  paid_at DATETIME NULL,
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT pk_payment PRIMARY KEY (id),
  CONSTRAINT fk_payment_invoice FOREIGN KEY (invoice_id)
  REFERENCES invoice(id) ON DELETE RESTRICT ON UPDATE CASCADE,
  CONSTRAINT fk_payment_method FOREIGN KEY (payment_method_id)
  REFERENCES payment_method(id) ON DELETE SET NULL ON UPDATE CASCADE,
  CONSTRAINT chk_payment_amount CHECK (amount > 0),
  INDEX idx_payment_invoice (invoice_id),
  INDEX idx_payment_method (payment_method_id),
  INDEX idx_payment_status (status),
  INDEX idx_payment_transaction_id (transaction_id)
);
```

dunning_event

Tracks automated payment retry and reminder events for overdue invoices. Each row is one step in the collections sequence.

Column reference

Column	Type	Notes / Constraints
id	CHAR(36)	Primary key — UUID
invoice_id	CHAR(36) FK	References invoice — CASCADE on delete
event_type	ENUM	email_reminder retry_charge suspension_warning final_notice
attempt_number	INT	Sequence number of this retry — CHECK > 0
scheduled_at	DATETIME	When the event is due to execute
sent_at	DATETIME	Actual dispatch time — nullable until sent
status	ENUM	pending sent failed skipped

CREATE TABLE statement

```
CREATE TABLE dunning_event (
  id          CHAR(36) NOT NULL DEFAULT (UUID()),
  invoice_id  CHAR(36) NOT NULL,
  event_type  ENUM('email_reminder','retry_charge',
                  'suspension_warning','final_notice') NOT NULL,
  attempt_number INT NOT NULL DEFAULT 1,
  scheduled_at DATETIME NOT NULL,
  sent_at     DATETIME NULL,
  status      ENUM('pending','sent','failed','skipped')
              NOT NULL DEFAULT 'pending',
  CONSTRAINT pk_dunning_event PRIMARY KEY (id),
  CONSTRAINT fk_dunning_invoice FOREIGN KEY (invoice_id)
    REFERENCES invoice(id) ON DELETE CASCADE ON UPDATE CASCADE,
  CONSTRAINT chk_dunning_attempt CHECK (attempt_number > 0),
  INDEX idx_dunning_invoice      (invoice_id),
  INDEX idx_dunning_status      (status),
  INDEX idx_dunning_scheduled_at (scheduled_at)
);
```